

lateau at around 4 K and stays unchanged at lower temperatures, indicating a transition to a plateau-like state in Region I. Low-energy spin variations are reflected in this plateau. The ensuing increase in the

A shift to slower spin dynamics is indicated by the relaxation rate in Region-II as the temperature drops, most likely as a result of the formation of short-range spin correlations. As the temperature falls below 15 K, the relaxation rate increases dramatically, signifying the transition from the paramagnetic phase to a possible quantum spin-liquid (QSL) phase.

This behavior is typical of systems having QSL-like properties, in which the spins fluctuate even at low temperatures rather than ordering conventionally. Other quantum spin-liquid possibilities have shown similar crossovers.